

Biological Database

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Introduction to Biological Data

Introduction

Biological data refers to any information collected from the living world. This includes:

- DNA, RNA and protein sequence data
- Genetic, population and ecological information
- Experimental results and computational analyses

With modern biotechnology and sequencing tools, the amount of biomolecular data is exploding. To store, manage and share this information globally, scientists use biological databases.

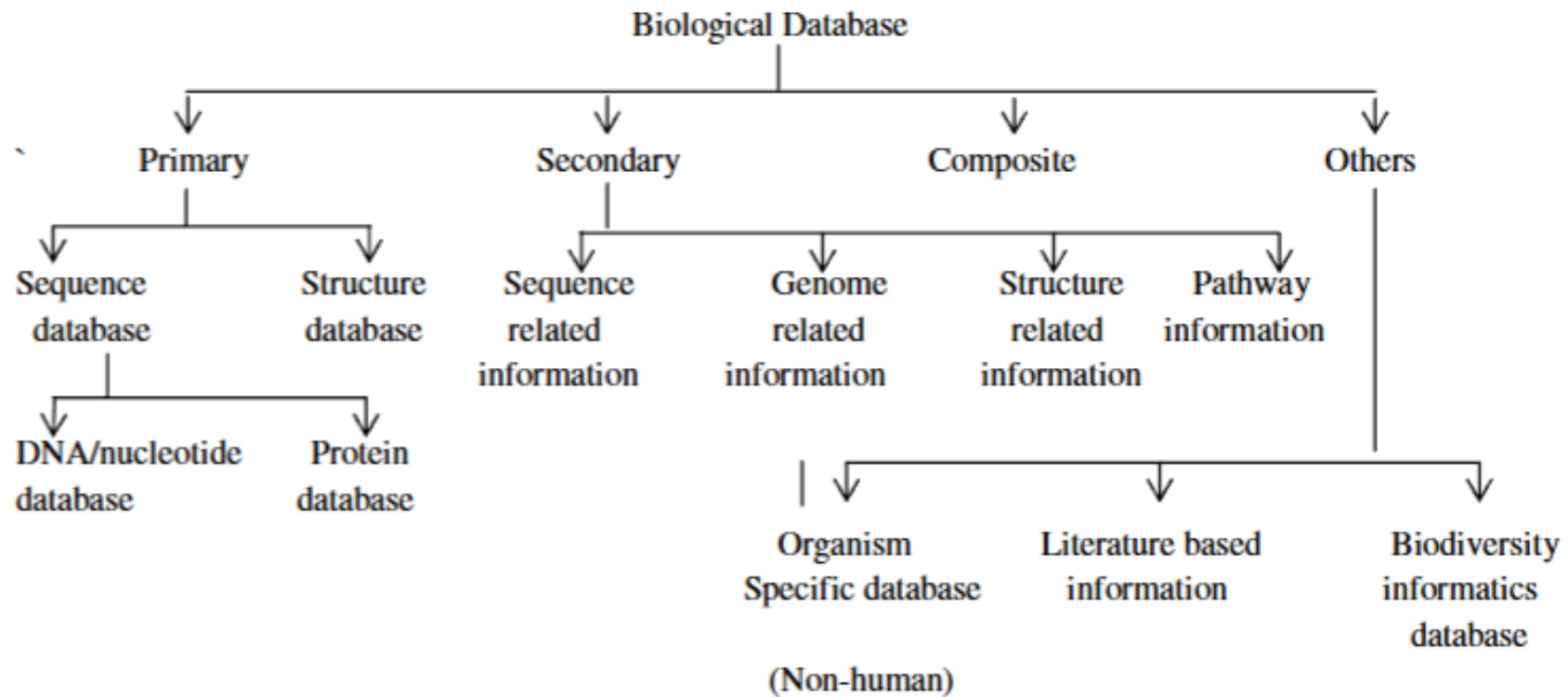
What Are Biological Databases?

A biological database is a structured, organized collection of biological information.

- Each database entry may include:
- Nucleotide or protein sequence
- 3D structure
- Gene function
- Organism source
- References and submitter information

Databases allow easy access, update, comparison and computational analysis of biological information.

Classification



Major Categories of Biological Databases

A. Bibliographic Databases

- Contain scientific literature.

Example: [PubMed](#)

B. Taxonomic Databases

- Contain classification and naming of organisms.

Example: NCBI Taxonomy

C. Nucleic Acid Databases

Store DNA and RNA sequences

Examples:

GenBank

DDBJ

[EMBL-EBI](#)

Major Categories of Biological Databases

D. Genomic Databases

- Contain complete genomes and their annotations.

Example: Ensembl Genome Browser

E. Protein Databases

- Contain protein sequences and functional information.

Examples: UniProtKB / Swiss-Prot, PIR

F. Protein Families, Domains and Functional Sites

Store conserved protein domains, motifs and families.

Examples:

Pfam

PROSITE

G. Enzyme/Metabolic Pathway Databases

Contain enzyme functions and biochemical pathways.

Example: KEGG

Types of Biological Databases (Based on Data Nature)

A. Primary Databases

Contain raw experimental data (sequences, structures).

Examples:

GenBank (DNA)

[PDB](#) (protein 3D structure)

UniProtKB/Swiss-Prot (curated protein sequences)

Types of Biological Databases (Based on Data Nature)

B. Secondary Databases

Analyzed or derived information from primary data.

Examples:

Pfam (protein families)

PROSITE (motifs)

HSSP (secondary structure alignment)

Types of Biological Databases (Based on Data Nature)

C. Composite Databases

Combine multiple primary databases into one platform.
Examples:

UniProt (PIR + Swiss-Prot + TrEMBL)

INSD (GenBank + EMBL + DDBJ)

nrdb (non-redundant sequence database)

Types of Biological Databases (Based on Data Nature)

D. Other Databases

Focused on specific organisms or literature.

Examples:

PubMed (literature)

Organism-specific databases (yeast, bacteria, insects)

Biodiversity informatics databases

Key Primary Databases and Their Roles

GenBank

- NCBI-managed repository of nucleotide sequences
- Accepts submissions via BankIt or Sequin
- Retrieval via Entrez or FTP

DDBJ (Japan)

- Accepts global submissions and shares data with GenBank & EMBL

Key Primary Databases and Their Roles

EMBL-EBI (Europe)

- First nucleotide sequence database
- Stores data on genes, proteins, pathways, structure, expression

Ensembl

- Automated genome annotation platform
- Provides gene locations, variations, and regulation data

PDB (Protein Data Bank)

- Repository of 3D protein and nucleic acid structures obtained from X-ray, NMR etc.

Secondary Databases

ProSite

- Protein families, motifs, and functional sites
- Manually curated patterns and profiles

Pfam

- Protein families represented as HMM profiles
- Contains Pfam-A (curated) and Pfam-B (automated)

REBase

Restriction enzyme information

OMIM

Human genes and genetic disorders

Each gene/disease assigned a unique six-digit number

Structure-Related Databases

DSSP: Secondary structure assignments for PDB entries

HSSP: Homology-based structural information

FSSP: Similar protein structures

NDB: 3D structures of nucleic acids

CSD: Small molecule structures

Pathway Databases

- Provide molecular interactions, enzymes, compounds and reactions

Example: [KEGG](#)

Retrieval Systems for Biological Databases

Entrez (NCBI)

- Integrates sequences, genomes, structures, literature
- Cross-links between related data

SRS (EBI)

- Searches across 80+ biological databases

DBGET/LinkDB (Japan)

- Linked with KEGG
- Useful for pathway-based searches

List of Key Databases

- <https://docs.google.com/spreadsheets/d/1tklz5Sp2MD7WVY8ouyMQJLbn4ZuEsBs7we2oP6EqK04/edit?gid=1260536385#gid=1260536385>

Thank you